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Developing a playable simulation of markets and corporations

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1 Motivation

During Economy Week (Deutsch: Wirtschaftwoche), as an exercise in *business administration*, we *played* a game, where you were the board of a company manufacturing some arbitrary product and had to make decisions to increase sales, lower cost and generally beat out you competition. For me, the aspect of being able to come up with novel management ideas and compete with your friends really hooked me into the concept. This may sound hyperbolic, but during that week I genuinely woke up exited to go to school, not to actually learn stuff (and whatever) but just to play a few more rounds and (aspirationally) completely destroy my competition.

The game we got to play during Economy Week, in my opinion, had a tantalising hook but the interface was utterly hostile to non-directed interaction: it was pretty much an excel table! Anyway, I set off to make a better, deeper version of this simulation.

> “We do this not because it is easy, but because we thought it would be easy”
>
> **Not JFK.**

This is the running motto of this project.

2 Concept

The concept behind the game was originally to create a game where the company creates and runs a company that manufactures and sells a product while competing with there friends. Imagine a sort of Civilisation for digital entrepreneurs.

The player has to battle against forces from within and without. The player has to develop a product, attempting to target a specific segment of the market, they have to balance its price with the cost of development and manufacturing, all the while trying to beat out the competition.

3 Design

The game would work in 2 phases: the *decisions phase* and the *simulation phase*.

During the decisions phase the player would set the values for various decisions like how many production machines they choose to buy. Just like the what we played during Economy Week, the interface was a thinly veiled Excel table.

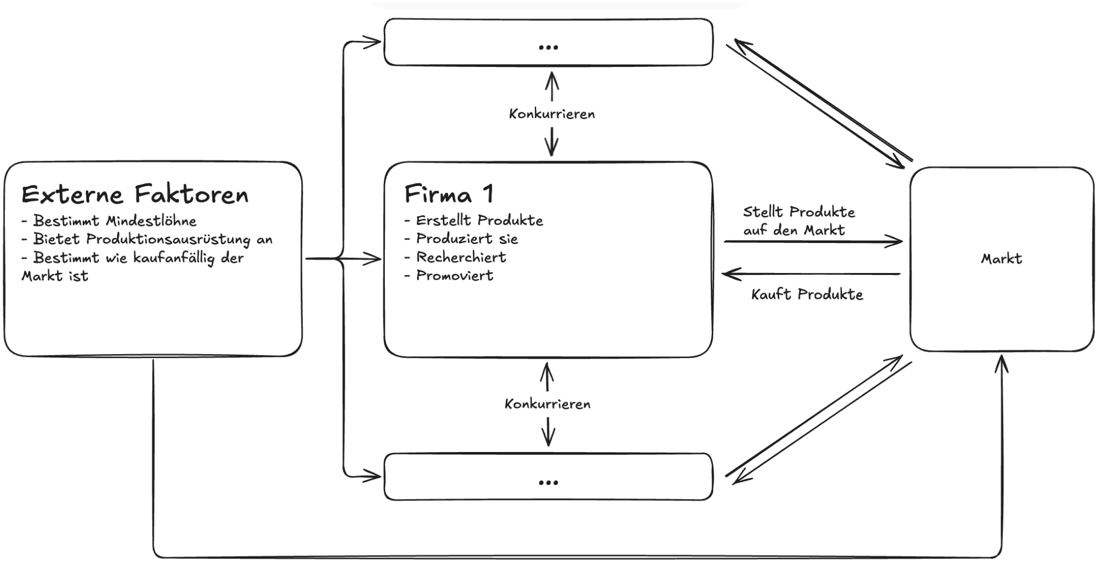

Figure 1: Diagram of factors influencing the player's decisions

After trying to "play" this for a few minutes I simply thought to myself: "This can't be fun, let alone would anyone care show this to their friends!", and so I gave myself a new blank canvas and actually tried thinking about what I wanted to make.

After a short break I took some time deisgning a proper user interface. As a "used to be avid gamer before school started taking up all of my free time" -er, I let myself be inspired by the games I *used* to play the most often.

One example the product designer. In early versions of the game, the product designer was simply a menu of sliders, all of which changed a certain stat in an unpredicatable and unrepeatable manner. This design was an unspectacular mix of boring, unintuative and intimidating. During early playtesting players repeatedly asked what each slider did and *why* it did it. Understanding what was actually happening was tremendously difficult and theoretical (like playing and Excel table).

Thinking of how to fix the above problems, I remembered Hearts of Iron 4. Hearts of Iron is a WW2 simulation game and it suffers from the same problem of making an Excel table fun to play. One of their early DLCs included a new tank designer in which the player chose specific components to put in their tank. This system decreases the abstractness of the content while still allowing for lots of creativity. Seeing the similarities between the 2 games I promptly adapted the system to my game.

The tank designer system did, however, require some major changes: the most important of which being that it needed a concrete product to be produced. At this stage in the games development, each company was developing a generic

Figure 2: The first UI I made for the game

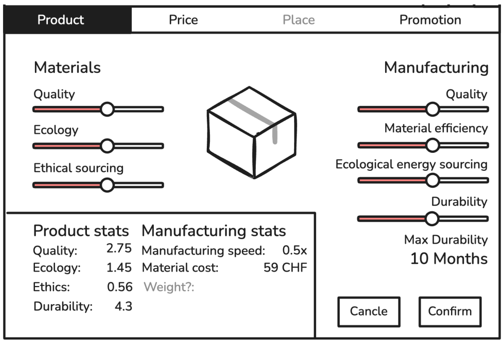

Figure 3: The original product designer

Figure 4: The tank designer from Hearts of Iron 4 (left) and from Wisim (right)

product. And so after (very short) deliberation, I picked coffee machines, for no deeper reason than there was a coffee machine sitting right in front of me.

4 Technology

When you start developing a piece of software of any kind, you have to pick a stack, that being: choosing which technologies you're going to build your project on top of. This choice is probably the single most important decision of any project (right after choosing what to build); getting this wrong can lead to headaches for months to come.

This project experienced two major rewrites where I swapped platform! The back-end, where the *real calculations* take place is written in "Go" (commonly referred to as Golang), for those who aren't familiar with it, it's like if Python (Tygerjython) and C [1] had a baby; it has a simple syntax like python with 90% of the performance of C. The front-end (what the user actually sees) was originally written in HTML, CSS & JavaScript using "Wails" to connect the front- and back-ends. A few months into development I was experiencing too many issues with this setup: random crashes, tight coupling [2] and unexplainable bugs. This frustrated me to such an extent that I just stopped working on the project for multiple weeks. Eventually I decided to completely rewrite the front-end in Godot, a free game engine; the problem? I didn't know Godot. The smallest of changes required hours of reading the docs and struggling around.

In the end I came back to my senses and tried developing the front-end in JavaScript, this time learning Web-Sockets . As any developer knows, the 3rd time you write something is always better than the first two times. And so I stuck with this right until the end.

5 Iteration

6 Simulation

When attempting to simulate a consumer market, there are quite a few approaches you could choose between. The vast majority of models fall into 2 categories:

1. Statistical models: models where the simulation uses regressions and other statistical analysis tools to determine outcomes.
2. Agent-based models: these models choose to, instead of simulating the whole market at once, instantiate individual *agents* that have the ability to

[1] The "C" programming language is the basis of most modern software infrastructure, being used by every operating system. It's very performant, however it has very few features, making the developer have to develop many things from scratch

[2] Coupling in software refers to how when one part of the software is changed, another related part also has to be as well.

independently make decisions.

There are 2 main advantages to statistical models: they are simpler to build, as you can often base them around simple regressions while still returning convincing results, and performance, unless the implementer makes grave mistakes, statistical models almost always outperform (in terms of speed) agent-based models of the same caliber.

For example, the cult classic city builder SimCity 4 was able to simulate regions with populations up to 100 million with very minor performance issues, [3] while over 20 years later, games like Cities: Skylines 2 are still plagued by performance issues from simulating just 100'000 agents on modern multi-core hardware.

The biggest downside of statistical modelling for this project is flexibility. An agent-based model can simulate virtually anything, should the designer choose to, without necessarily major changes. This was the deciding factor for me, as the development of the game would take time, during which I would inevitably want to rework aspects of the simulation to optimise certain aspects.

6.1 How it works

The game works in 2 phases: the decisions phase and the simulation phase. During the decisions phase players can analyse the data provided to them and ajust their companies, designing new products, hiring workers, purchasing production machines, etc. When every player is ready, the simulation phase begins, during which the decisions made in the previous phase are executed.

Products

Players will develop many products during the course of the game. The products are made up of various comoonents, these comoonents along with the research levels of the company impact the properties of the product, for example using wooden construction will make you product slightly more ecological whereas using plastic will make it less so.

Players have to balance quality, ecology, durability and ethics with the manufacturing difficulty, design costs and price. This forces players to attempt to target a specific customer base lest they lose agaist someone who does. For example: the product that objectively has the best stats may be so expensive to produce that it

[3] This Kotaku article from 2014 shows off a city of over 100 million inhabitants, the Guinness world record, without any performance problems.
Link to article: https://kotaku.com/simcity-megacity-has-over-100-000-000-million-people-li-1629131314;
World record: https://www.guinnessworldrecords.com/world-records/418977-largest-simcity-4-city-region

has to be sold for multiple thousands of Francs, making it out of reach for most of the consumer base.

Production

Along with developing a product, companies of course have to produce them. To do so they need 3 things: production machines, employees and money. Every product has a so called ”production cost”, this refers to how difficult the product is to produce; a production machine will produce more products with low production cost per month than products with a high cost all else being equal. Every machine has a specific ”production capacity”, as you might expect, this refers to how much ”production” can be produced per month, thus how many of a specific product.

The second ingredient of production is employees, without them machines won't run. The key gameplay impact of employees is that as well as their salaries, the product of their skill and motivation determines how efficiently they work. If a machine has highly skilled, motivated workers it will produce even more than its listed production capacity. This encourages players to retain their best workers and treating them well. If they are over-worked they can burn out and become virtually useless. However, if you fire them, they might be snatched up by your competition, with them perhaps profiting off years of investment.

Last but not least is money. As with any physical good, they have to be made out of something and this something costs money. In fact during hard economic times the cost materials can increase dramatically, which, depending on the buisiness strategy, can impact the company's profits.

Consumers

The customer is simulated as a bundle of different characteristics, preferences, and needs; these all combine to produce unique behaviors for each individual agent. Taking inspiration from Economy Week, each customer has the following properties: [4]

[4] The names provided the table may not be accurate to the corresponding implementations found in the code to simplify the explanation, as the implementation contains subtle differences for performance and technological reasons.

Property	Type	Description
Base Need	Number	The amount of products a customer wants. When their existing products wear out, they buy more.
Owned Products	List[number]	Keeps track of the remaining durability of owned products.
Purchasing Threshold	Number	How attractive a product has to be for the customer to buy it.
Savviness	Number	How well informed the customer is.
Loyalties	List[number]	How loyal the customer is to a certain company.
Loyalty Factor	Number	How much the customer's loyalties influence their
Preferences	Dictionary[number]	Which aspects of a product impact the customer's decisions the most. For example, a customer could value low prices more than they do the quality of the product.

Originally the preferences were determined by sampling a *weighted normal distribution*; however, after some playtesting and reading The Tipping Point by Malcom Gladwell, I reworked the generation of customers by instead creating *archetypes* and slightly modifying their preferences randomly.

Each of the archetypes has some preferences boosted and others diminished. For example, "the enthusiast" will not shy away from buying an expensive product, even multiple, if they perceive it to be unique and high quality. On the other hand, "the Everyman" will prefer a simpler, cheaper, long-lasting product.

archetypes have not yet been fleshed out

6.2 The Marketing Phase

In this phase, the marketing campaigns of companies are simulated. Each marketing campaign has a style, a quality and a quantity.

The style is determined by the player, which can influence the perception of the product. This can be used to round out the *less* attractive parts of the product. For example, if a player had a high quality product that they wanted to market to a more eco-focused audience, instead of having to undertake the difficult and expensive process of actually producing in an environmentally friendly manner, they can just focus on promoting the product as eco-friendly. [5]

[5] This technique is often labelled as Greenwashing. The most famous example is Volkswagen's "clean diesel" emissions scandal in 2015 where Volkswagen programmed diesel engined cars to run efficiently when they were being tested while reverting to "dirty" operation when driven on the road.

The quality of a campaign influences how much the perception of the product is changed. This is mostly determined by the skill and motivation of the companies marketing staff.

Lastly, the quantity of a campaign, expressed in a monetary amount, determines who even sees the campaign. A high quantity of marketing means there is a high chance that people know about a product. A campaign could theoretically be extremely effective, however if no-one sees it, it isn't actually successful since there are still few people who even know about the product's existence.

All of these factors are mixed in with each customer's "savviness" stat. People with more savviness are generally more informed on a market and thus have a higher chance of knowing about a product, however they are also less effected by marketing and thus perceive it more objectively. Other people with lower savviness will easily be swayed by effective marketing.

6.3 The Purchasing Phase

After the effects of marketing are evaluated, if the customer needs a product, it will evaluate the dot product of its preferences and the product's properties (the decision factors) for each product.

$$\text{decision factors} = \begin{pmatrix} \text{quality} \\ \text{...} \\ \text{durability} \end{pmatrix} \cdot \begin{pmatrix} \text{quality preference} \\ \text{...} \\ \text{durability preference} \end{pmatrix}$$

The customer then checks if the product is still available; if so, they try to buy enough to fulfill their needs.

7 Actually making it fun

Further reading: https://en.wikipedia.org/wiki/Volkswagen_emissions_scandal